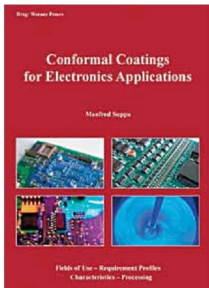


Examination of Conformal Coated Board and Functional Testing

Electronic assemblies are conformal coated worldwide as a protection against environmental conditions. These assemblies should be carefully inspected and tested for functionality after coating to prevent failures at system level testing stage



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When looking at the coating of an assembly, one must distinguish between the testing, examination of the coating process, the coating result, functional testing of the assembly and qualification testing of a coating.

The coating result can be examined, for example, in accordance with the rules of J-STD-001 and IPC-A-610, or also in accordance with the Workmanship Standard of NASA, NASA-STD-8739-1.

Tests carried out after coating of the assembly

Besides fulfilling the underlying technical specifications, the following is generally demanded for a coating in accordance with J-STD-001 and NASA-STD-8739-1:

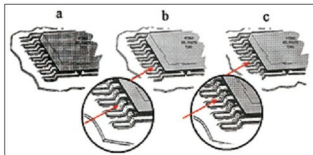


Fig. 1: Bubbles in a conformal coating: (a) preferred—uniform coating result without any visible bubbles, (b) acceptable—small bubbles not bridging any leads and (c) unacceptable—excessive formation of bubbles

1. The coating shall be uniform and even in colour, layer thickness and surface; minor pull back (edge loss) from sharp edges and angles is permissible unless otherwise specified in the technical specification

2. The coating shall properly adhere to all coated surfaces

3. Thorough drying or curing

4. Homogeneous coating film

5. Surfaces may be coated only in accordance with the respective specification

6. The coating film must be free from bubbles which have an impact on the protection provided by the coating

7. The coating film shall be free from voids, bubbles or inclusion of impurities that are likely to expose electric conductors and/or have an influence on the potential distances

In addition, IPC-CC-830B requires that the coating film shall be free from harmful substances, bubbles, pinholes and/or voids, wrinkle formation, crack formation and delamination.

The conformal coating result shall be smooth and even, homogeneous, transparent and tack-free at ambient conditions. As far as the forma-

tion of bubbles in the coating film due to a specific application method is concerned, the IPC handbook IPC-HDBK-830 gives a practical hint. Bubbles shall be avoided by all means in the dried coating film; since this is not always possible for different reasons, it

recommends the following:

“Bubbles are generally acceptable provided their size does not exceed 50 per cent of the distance to the conductor from the spot where they occur, and they do not expose any conductor, bridges or adjacent conductor surfaces.”

The NASA-STD-8739-1 released in 1999 gives visual assistance (see Fig. 1) by describing the frequency of bubble formation.

According to NASA-STD-8739-1, the bubbles of Fig. 1(b) shall “not exceed 0.76 mm.” However, given the progress in miniaturisation since the standard was published in the year 1999, this statement has to be considered outdated. The IPC-A-610D view on bubbles is as follows: “The inspection shall be done with the naked eye. If this inspection clearly confirms the suspicion of bubble formation, the parts in question shall be viewed with a maximum of 4x magnification (e.g., by a 3x ESD-protected loupe), according to IPC-A-610D, 10.5.2.2. This magnification is based on the description of J-STD-001D. Coating materials that contain a fluorescent pigment may be examined under black light to verify the